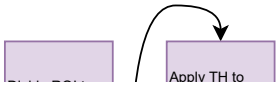
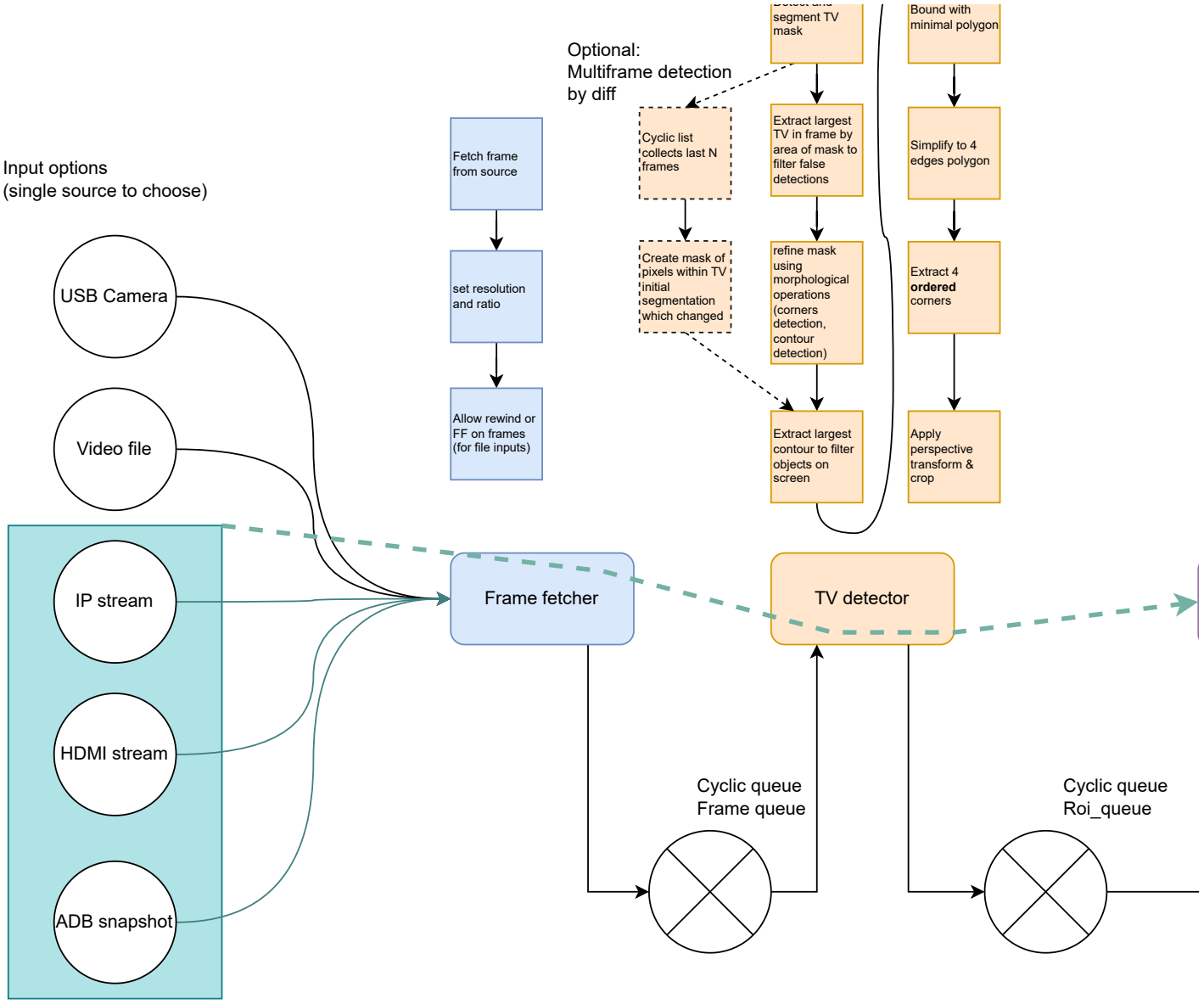
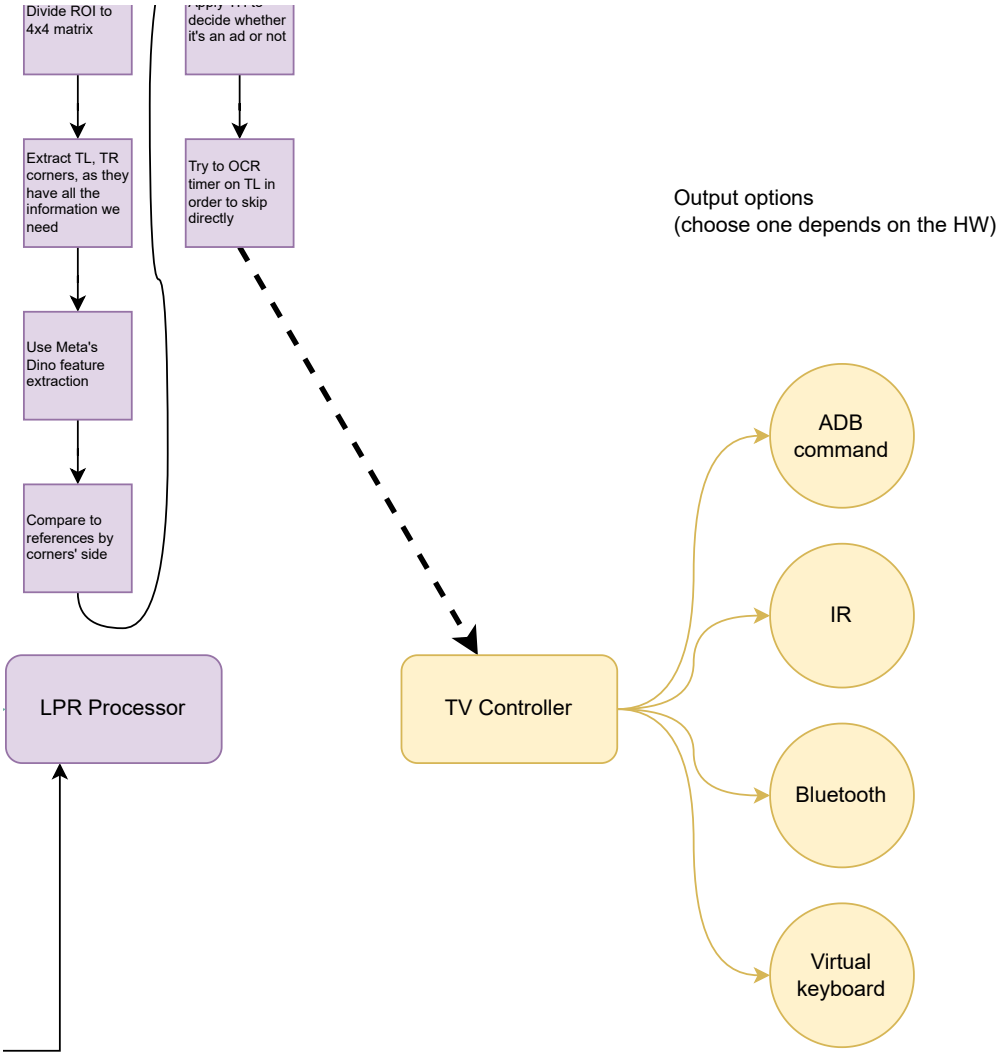

Detect and


Apply TH to


Input options
(single source to choose)
USB Camera
Video file
IP stream
HDMI stream
ADB snapshot
Fetch frame from source
set resolution and ratio
Allow rewind or FF on frames (for file inputs)
Frame fetcher
Optional:
Multiframe detection
by diff
Cyclic list collects last N frames
Create mask of pixels within TV initial segmentation which changed
segment TV mask
Extract largest TV in frame by area of mask to filter false detections
refine mask using morphological operations (corners detection, contour detection)
Extract largest contour to filter objects on screen
Bound with minimal polygon
Simplify to 4 edges polygon
Extract 4 ordered corners
Apply perspective transform & crop
TV detector
Cyclic queue
Frame queue
Cyclic queue
Roi_queue


Divide ROI to 4x4 matrix
decide whether it's an ad or not
Extract TL, TR corners, as they have all the information we need
Try to OCR timer on TL in order to skip directly
Use Meta's Dino feature extraction
Compare to references by corners' side
Output options
(choose one depends on the HW)
LPR Processor
TV Controller
ADB command
IR
Bluetooth
Virtual keyboard